

CAPTURING PHOTONS

PHOTON COLLECTION

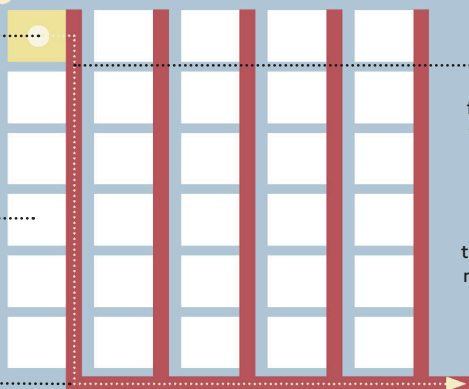
Each capacitor accumulates a charge, which is proportional to the intensity of light at that point.

PIXEL

Each pixel is represented by a capacitor.

HORIZONTAL SHIFT REGISTER

Transfers charge to processor for output.



CIRCUIT

The control circuit forces each capacitor to transfer its charge to its neighbor.

The final capacitor in a column then transfers its charge to the horizontal shift register, which moves it to a processor.

CHARGE-COUPLED DEVICE

A charge-coupled device (CCD) is a circuit of linked capacitors, which store and transfer electrical charge. CCD image sensors convert light into electrical signals and are important components in digital imaging. Pixels are represented by semiconductor capacitors that free electrons when photons are absorbed, accumulating electrical charge proportional to light intensity at that point. A control circuit transfers charges through a capacitor array, converting them into a readable signal.